



Gold-layer NULL-rate stats — identity resolution problem

Reference data for Part 3 (Platform Design). These statistics motivate the gold-layer redesign — particularly the `dim_user` work. Use them as factual grounding when proposing the structure that replaces them.

Background

Office Timeline's gold layer currently has three `dim_pes_*` tables intended to provide a unified view of subscriptions, billing, and users. In practice, the silver-to-gold transform fails to resolve user identity for the majority of records — most rows in the gold dimensional tables either have NULL on the cross-system join keys or fail to round-trip back to the application layer.

The statistics below illustrate the scale of this gap. They are produced from a snapshot taken in mid-April 2026.

`dim_pes_user`

Intended to be the canonical user dimension joining PES users to HubSpot contacts. One row per resolved PES user.

Total rows: 327,841

Column	NULL count	NULL rate
<code>pes_user_guid</code> (PK)	0	0.00%
<code>email_normalized</code>	18,217	5.56%
<code>hubspot_contact_id</code>	289,043	88.16%
<code>hubspot_company_id</code>	304,506	92.88%
<code>first_seen_date</code>	0	0.00%

Column	NULL count	NULL rate
<code>is_admin</code> (subscription admin flag)	0	0.00%
<code>country</code>	41,672	12.71%

Reading this: of the 327,841 users known to PES, only 38,798 (11.84%) have a resolvable `hubspot_contact_id`. The remaining 88% are either users PES knows about but HubSpot doesn't, or — much more commonly — users who exist in both systems but whose join failed because of email case sensitivity, late-arriving HubSpot contacts, or admin email changes that broke the link after initial resolution.

dim_pes_subscription

One row per PES subscription, latest state.

Total rows: 47,238

Column	NULL count	NULL rate
<code>subscription_id</code> (PK)	0	0.00%
<code>subscription_tracker</code>	38,114	80.69%
<code>administrator_pes_user_guid</code>	2,891	6.12%
<code>administrator_hubspot_contact_id</code>	41,604	88.07%
<code>customer_email_normalized</code>	1,148	2.43%
<code>quantity_current</code>	0	0.00%
<code>start_date</code>	0	0.00%
<code>expiry_date</code>	312	0.66%
<code>is_active_derived</code>	0	0.00%

Reading this: the subscription table itself is well-populated, but the subscription-to-HubSpot join goes through `administrator_hubspot_contact_id`, which has the same ~88% NULL rate as `dim_pes_user`. This means any analysis that requires "which subscriptions belong to which HubSpot company" is impossible at gold for ~88% of subscriptions.

`subscription_tracker` is NULL for 80.7% of subscriptions because only post-ABL subscriptions populate this field (ABL launched November 2025 and most large

subscriptions remain on the legacy product-key model).

dim_pes_billing

One row per billing event.

Total rows: 89,156

Column	NULL count	NULL rate
<code>order_id</code> (PK)	0	0.00%
<code>subscription_id</code>	12,481	14.00%
<code>customer_email_normalized</code>	7,132	8.00%
<code>buyer_email_normalized</code>	18,723	21.00%
<code>license_manager_email_normalized</code>	32,099	36.00%
<code>pes_user_guid</code> (resolved buyer)	49,036	55.00%
<code>hubspot_contact_id</code> (resolved buyer)	81,222	91.10%
<code>discriminator</code>	4,458	5.00%
<code>gross_amount</code>	0	0.00%

Reading this: even more than half of billing events fail to resolve to a PES user GUID, and 91% fail to resolve to a HubSpot contact. This means questions like "which marketing source did this paying customer originally come from?" cannot be answered for the vast majority of revenue.

A concrete illustration: a single enterprise customer

A spot-check on one large enterprise customer (a 1,000+ employee company that has been an OT customer for 8+ years):

Layer	Records visible
<code>silver.pes_subscriptions</code> (latest state, all subs for this customer)	849 seat assignments across active subscriptions
<code>silver.hs_otl_contacts</code> (contacts with email at this customer's domain)	612

Layer	Records visible
<code>gold.dim_pes_user</code> (rows for this customer's PES user GUIDs)	53
<code>gold.dim_pes_user</code> (rows where <code>hubspot_contact_id</code> is non-NULL)	9

Of 849 seat assignments visible in silver, only 53 (6.2%) make it through to a row in `dim_pes_user`. Of those 53, only 9 have a resolved `hubspot_contact_id`. So for the largest customers, where seat-level analysis matters most, the gold layer surfaces ~1% of the underlying data.

This is the kind of customer where a renewal conversation, expansion play, or churn-risk analysis would benefit most from a unified user view. Today, that analysis cannot be done at gold — it has to be done by reaching into silver and reproducing the identity resolution logic ad-hoc, which is exactly what `reporting.view_funnel_dashboard` does.